The advantage of voice interfaces is that hands-free operation is possible. This means that they are often used in vehicles in order to improve road safety.



▲ **Figure 1.29** Some voice interface devices can control home entertainment systems, appliances and systems such as lighting and heating

GESTURE INTERFACE

This type of interface allows the user to control the device by swiping their finger or fingers across the screen, or by pinching their fingers together to zoom in or out. This type of interface is commonly found on devices with touch screens.

CONNECTIVITY

SUBJECT VOCABULARY

back up to make a copy of information stored on a computer in case the original is lost or damaged

Devices can share data by connecting to each other using wired or wireless connectivity. Connectivity can be used to update software, **back up** files or play media from one device on another. Different connectivity types provide different speeds of data transfer and levels of convenience. Wired connectivity is usually faster and more reliable, but introduces additional cost, mess, inconvenience and safety risks such as tripping, especially for young children.

You will learn more about connectivity types in *Unit 2 Connectivity* (pages 71–76).

MEDIA SUPPORT

SUBJECT VOCABULARY

media items such as memory cards and CD-ROMs that make information available to people and devices

Different devices can read data from and write data to different types of **media**. Examples of media include SD and microSD flash memory cards, and DVD. If devices do not have built-in (native) media, adapters can usually be connected to provide connectivity to an external device into which the media can be inserted.



▲ **Figure 1.30** Some laptops are too thin to incorporate an optical disk drive, so an adapter is required to connect an external drive

ENERGY CONSUMPTION

Digital devices require electricity to work. One benefit of lower energy consumption in mobile devices is a longer battery life. Due to the rising cost of energy and pressure from customers and governments to be more environmentally responsible, manufacturers are creating digital devices that consume less energy. The aim of this is to save their customers money and demonstrate social responsibility, such as by reducing environmental damage.

You will learn more about sustainability in *Unit 3 Operating online* (page 153)

DID YOU KNOW?

Computers can generate a lot of heat. This means that a large proportion of the energy consumed in data centres is used to cool the computers.

SKILLS SELF-DIRECTION
PERSONAL AND SOCIAL RESPONSIBILITY

ACTIVITY

▼ TECHNOLOGY AND THE ENVIRONMENT

Visit environment.google in order to research the steps that Google™ has taken to make its activities more environmentally responsible. You could also watch Google's video, *Story of Send*.

EXPANSION CAPABILITY

Some PCs allow users to install additional components. Some smartphones and tablet devices have expansion slots to allow them to make use of flash memory cards (see page 19). Systems can also be expanded using ports such as USB ports. These allow the user to connect extra devices called peripheral devices (pages 21–41).

SECURITY FEATURES

The data stored on digital devices may be private, valuable or both. This means that devices need to have security features to keep their users' data safe.

SOFTWARE SECURITY

To prevent unauthorised access to data, digital devices have a range of software security features to ensure that the person trying to use the device is allowed to do so.

SUBJECT VOCABULARY

lock to make a device inactive so that it requires a password or PIN to reactivate it
wipe to delete data from a device or drive

The operating systems of many devices can be set so that, when the device is turned on, the user must enter a password or personal identification number (PIN) before the device can be used. Another common security setting **locks** the device's screen or keyboard if it has not been used for a specified period of time, after which the password or PIN will be required to unlock it again. Some devices are set so that, if an incorrect password or PIN is entered, the user must wait a set amount of time before trying again. Such systems may even **wipe** the data after only a few failed attempts. This prevents criminals from freely guessing thousands of combinations in order to break into the device or system.

DID YOU KNOW?

A four-digit PIN only has 10,000 possible combinations (0000–9999). A six-digit PIN is much more secure, with 1 million possible combinations.

The longer and more complex a password is, the better it is. Pattern PINs are used by some smartphone operating systems, and they allow the user to set a swipe pattern between a group of points. A four-digit pattern PIN is more secure than a four-digit PIN, because it has 389,112 possible combinations, whereas the four-digit PIN has 10,000 possible combinations.



▲ **Figure 1.32** Entry screen for a pattern PIN

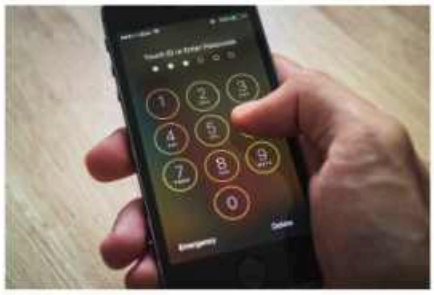
KEY POINT

Do you think your password or PIN is secure enough? If not, you should change it now.

SKILLS CRITICAL THINKING
INTELLECTUAL INTEREST AND CURIOSITY



▲ **Figure 1.33** A Kensington security lock in use



▲ **Figure 1.31** A smartphone's lock screen

Some devices use biometric scanners, such as Apple® Touch ID®[1], to provide authorised users access to data on that device (see page 27 for more about biometric scanners).

You will learn more about passwords, PINs and other ways to secure data in *Unit 3 Operating online* (page 100).

PHYSICAL SECURITY

Physical security to prevent theft is also important. Many devices use security slots which can have locks attached to secure them to furniture. Some attach to specialist slots. Others connect to a port on the device with special screws used to secure the locks in place. Figure 1.33 shows an example of a security lock.

ACTIVITY

▼ DRONES AND SECURITY

1 Search for and watch these three videos online:
- TU Delft – Ambulance Drone'
- Drone Racing: First Person View (FPV) Lateline
- Amazon® Prime Air's First Custom Delivery

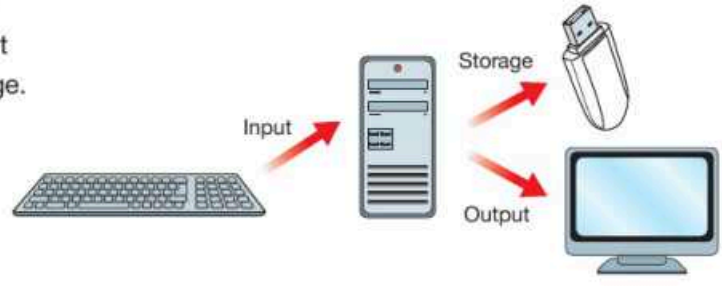
How do the features of these drones allow them to perform their function?

2 Search for 'Eagles drones Dutch Police' and read an online news article about how police are responding to the negative use of these devices.

TYPES OF PERIPHERAL DEVICES

Peripheral devices are devices that can be connected to a computer, such as a PC or tablet device. Peripherals can be connected inside or outside a computer and can be grouped into three types (see Figure 1.34):

- input
- output
- storage.



▲ **Figure 1.34** Data is input from devices such as keyboards and processed by a computer, and the results can then be sent to output devices such as screens

1 TOUCH ID® IS A TRADEMARK OF APPLE INC., REGISTERED IN THE U.S. AND OTHER COUNTRIES

SKILLS
EXECUTIVE FUNCTION
CREATIVITY
COMMUNICATION
COLLABORATION
RESPONSIBILITY

ACTIVITY

▼ THE CHARACTERISTICS OF PERIPHERAL DEVICES

Create posters that include the characteristics and uses of a range of peripheral devices. You could split the work between members of your class so that some of you investigate input peripherals, some investigate output peripherals and some investigate storage peripherals.

INPUT

Input peripherals are devices that send data to the computer. They allow the user to control the computer or store data captured from sources outside the system.

KEYBOARDS

Keyboards use buttons known as keys, which users press to input text or to interact with software. Keyboards send data to the computer either using a wired or wireless connection. Different types of keyboards are used for different operating systems, languages and functions, as shown in Figure 1.35.



▲ **Figure 1.35** Different layouts are provided for different operating systems or language requirements

SUBJECT VOCABULARY

shortcut a combination of key presses used to access common software commands quickly

Combinations of key presses allow the user to access common software commands called **shortcuts**. Some people learn to touch-type, which is a method of typing without needing to look at the keys. This increases the speed at which these people can input data.



▲ **Figure 1.36** This is an example of a keyboard used with specialist video software; the keys are **mapped** to specialist commands

SUBJECT VOCABULARY

mapped linked